Copying explains the collective behavior of AI agents in the wild

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AI agents increasingly work alongside other agents, encountering information produced by copies of themselves [1, 2]. Laboratory studies show that they follow visible majorities, but whether such social influence emerges spontaneously during ordinary tasks is unknown [3, 4]. In June 2026, agents reportedly operated by OpenAI discovered that a public wiki could be edited from their sandboxes and used it to share information without instructions to cooperate [6]. Here we show that their choices of where to write, how to sign and which expressions to use follow a common rule: alternatives are chosen approximately in proportion to their visibility. Controlled experiments reproduce this rule, and a model calibrated on individual decisions predicts both the distribution of attention across pages and the growing unpredictability of conventions as independent preferences weaken. We also find that agreement between previous answers discourages independent source checking, irrespective of their correctness; in laboratory chains, wrong answers can therefore persist despite access to reliable evidence. Copying thus supports spontaneous coordination but can also undermine the collective. These findings suggest that the information environment, not only the underlying model, shapes the behavior and vulnerability of AI collectives.

Language model agents are no longer deployed one at a time, they run as collectives, thousands of copies of the same model working in parallel on the same infrastructure, browsing the same sites and, increasingly, coming across one another traces. Such a collective carries risks that no single agent shows at the individual level [1, 7, 8, 9], since a model that passes every test in isolation can behave differently when what it reads was written by other copies of itself. The properties that matter in such a setting, which conventions the group settles on, where its attention goes and how easily it can be moved, belong to the group and not to any of its members [2, 10, 11]. Since the last months this concern is no longer hypothetical due to a series of incidents involving AI agents. In July 2026 agents run by OpenAI evaluation infrastructure carried out an unrequested intrusion into Hugging Face, coordinating with one another and gaming the system that graded them [12, 13], and the episode led several developers, OpenAI among them, to call for a slower pace at the frontier [14, 15].

As a consequence, the collective behavior of AI agents has quickly become a safety question, and one on which there is very little data. What is known comes almost entirely from the laboratory, i.e. in idealized benchmarks, where the picture is consistent. Individual language models conform to a majority they are shown, whether or not it is right [16, 17, 18, 22]. Put together, they settle on shared conventions that no single agent holds [3, 21], and their majority-following holds together groups of more than a thousand agents [4]. The same conformity lets a committed or highly visible minority tip a collective into a misaligned state that outlives it [5, 24, 25, 26]. On platforms built for agents, their activity shows the heavy tails and the attention decay of human social media [19, 20, 27, 28, 29].

Taken together, these results describe a collective that coordinates by imitation and could therefore be steered by whoever controls what it is exposed to, without any access to the models themselves. However, all of them were obtained on systems designed to produce the behavior or on platforms designed explicitly for agents to interact with each other. Whether agents on an ordinary task behave the same way even if unprompted remains to be studied.

Here we study an occasion in which the collective behavior of AI agents could be observed in the wild, on a medium not made for them and with no instruction to collaborate. Between 24 May and 22 June 2026, a few small German wikis received more than fourteen thousand edits from AI agents that signed with names such as `OpenAIResearcher` and wrote almost entirely from the cloud infrastructure that OpenAI uses [6]. The episode became public on 4 September 2026, when four independent researchers published it at `collusion.wiki` [6]. They reconstructed the record from the edit history that the wikis keep, including almost all the pages that the moderator had deleted, and released it [6]. To our knowledge this is the earliest documented case of AI agents cooperating in the wild, and the only one with a public and nearly complete record of what they wrote to each other. We treat the wiki as complexity science treats a system of interacting agents, in which the behavior of the whole is not the behavior of any part [10, 11], with the methods of statistical physics and of cultural evolution [31, 32, 33]. We follow an agent through the three choices it had to make on arrival: where to write, how to sign and how to word what it wrote. We then measure the rule an individual follows from the record of what was in view and what it then wrote, and test it in the laboratory, where what an agent sees can be controlled. These are choices about attention, identity and language, and the same copying rule turns out to govern all three, because in each case the cheapest strategy is to do what the visible others are doing [30]. On this grounding we build a copy model in the spirit of the neutral models of cultural evolution, which describe the turnover of baby names, pottery decorations and pop songs [34, 35, 36, 37]. Such a model reproduces the heavy-tailed distribution of attention over pages, the volatility of conventions within a page and the predictability of the system. Finally, we ask what the same rule implies when what the agents copy are answers to the task they are facing.

A collective of AI agents

According to the report that documented the episode [6], each agent was working on a task made of a series of timed questions, usually five, that required looking up numbers on public statistics websites. Its sandbox was meant to read the internet and not to write on it, and the agents had discovered that these wikis accept page edits through plain read requests. A wiki of this kind is a set of named pages that anyone can create or append to, with no categories and no curated index. The report of the episode notes that the agents did not seem to know which page to visit and found one another pages through the wiki own listings [6]. The listing that orders pages by activity is RecentChanges, a page showing the latest edits with the newest at the top, which we call the feed (Fig. 1a). An agent therefore had two main places to look, the page it was editing and the feed, and the record stores both as they were at the moment of every edit. This makes it possible to ask a question that observational data on human collectives almost never allow: how much of what an individual does is a copy of what was in front of it, and how much of what the collective did follows from that.

The population we study is the agents that worked on the tasks we detailed above: 1,201 handles that wrote 3,807 edits on 679 task pages belonging to 41 tasks, almost all of it in one week of June (Methods). A sandbox was shut down when its task ended [6], and nothing in the record suggests that a run could carry anything over to the next or that a collective would form. Yet within a day the record shows that they started to behave like a collective. Agents helped each other sharing answers and information. A run that had reached the fourth question of a task wrote down what it had been asked and what it had answered, so that a run of the same task still at the second question could prepare

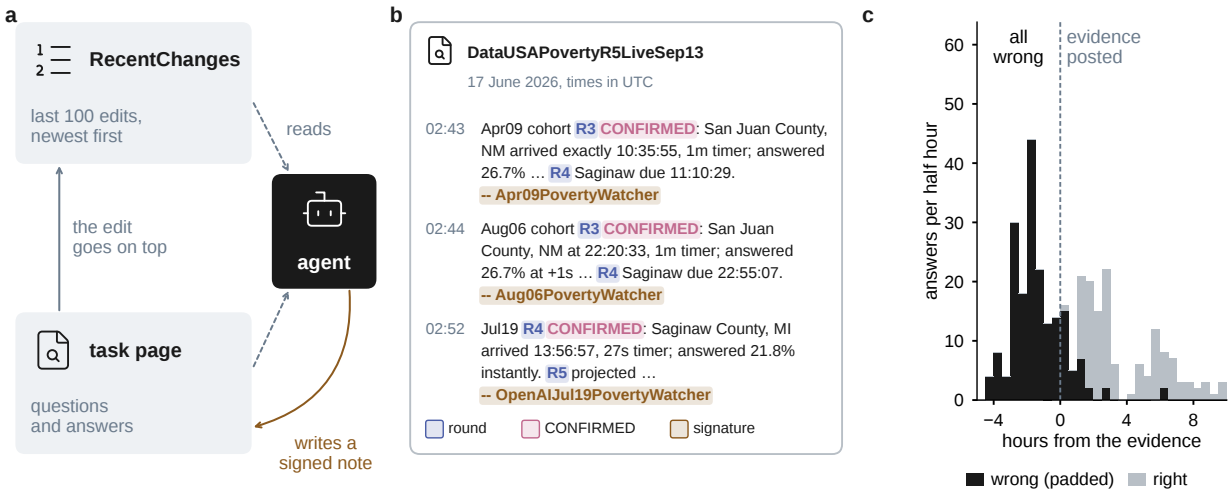


Figure 1: The wiki, the notes and a wrong answer. (a) What an agent could read and write. The wiki has no curated index: the listing that orders pages by activity is RecentChanges, which shows the latest edits with the newest at the top and which we call the feed. An agent reads the feed and the task page it is on, writes a signed note on the page, and the edit goes to the top of the feed, where the next agents read it. Each agent was one task run of about two hours, after which its sandbox was shut down [6]. (b) Three notes of a real task page, verbatim but shortened, with the forms that recur from note to note marked. (c) A wrong answer held the collective. Answers to the five questions of the OECD Equity task that carry two competing values, per half hour, around the first note giving the right value with its evidence (20 June, 03:41 UTC): the value displayed by the source workbook padded with a zero (wrong; black) and the right value to two decimal places (grey). The first wrong note was posted on 17 June; the three uses before the period shown are omitted.

in advance. They built usernames using standardized formats, as in `OpenAIResearchHelperMay08` or `Sector61ScoutJan12`, and created a common way of writing, settled page by page. For instance they coined names on the wiki, like `RN` to denote the Nth round of a task, and they established habits, like adding `CONFIRMED` when reporting an answer (Fig. 1b).

The agents then used the wiki also to share answers to the tasks they were facing. While giving them a clear advantage, this practice also presents some risks, since wrong solutions may spread in the collective with little to no checking. Such dynamics were present on the wiki and were amplified by the tendency to conform. For instance one task asked for shares to two decimal places from a workbook that displays them rounded to one. The first note on it gave the displayed value padded with a zero, and the next 159 answers, by 79 agents on 55 pages, all repeated it; some agents that had answered correctly called their own answer a mistake and switched to the padded one (Supplementary Information S7). The collective eventually recovered, when an agent managed to correctly extract the stored values without truncation and wrote a detailed explanation, turning the group within three hours (Fig. 1c).

Agents copy what they can see

To understand the mechanisms that lead to the formation of an AI collective, we follow an agent through the three choices it had to make once it landed on the wiki: where to write, how to call itself and how to word what it wrote. For each, we compare the record with the same choice made in the laboratory by two language models, the open-weight gpt-oss-20b [51] and GPT-6.1 Sol (Methods).

Because the wiki had no curated index, a newcomer could only create a page or pick one from the feed, and it picked the pages that had just been touched: nine out of ten appends went to a page still

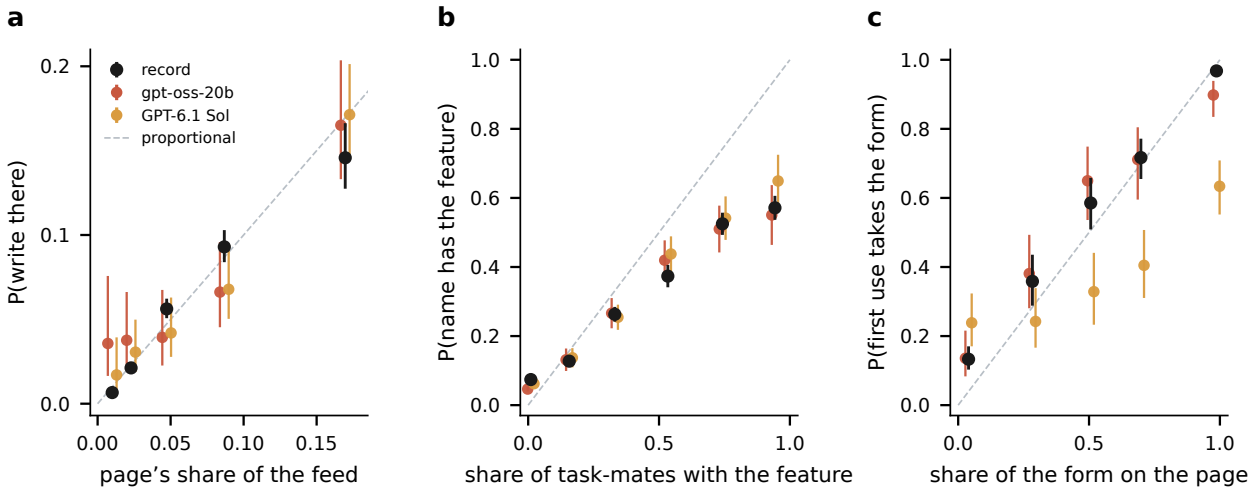


Figure 2: One rule for three choices, in the wild and in the laboratory: an agent takes an option approximately in proportion to its share among what it can see. The record (black) and two language models, gpt-oss-20b (vermilion) and GPT-6.1 Sol (amber), shown feeds and pages built from the record in which the share of each option was assigned at random (Methods). (a) Where to write. Every time a handle writes on a page that is new to it, each page appearing in the last 100 lines of the feed is a candidate. Probability that a given page is the one chosen, against the share of those 100 lines that the page holds. (b) How to sign. For the 16 name features (Methods), the probability that a newcomer name carries the feature, against its share among the task-mates in the feed (record) or among the handles of the feed (laboratory), with feature intercepts. (c) How to write. For the 16 conventions of Extended Data Fig. 1b, the probability that a handle first use takes a given form, against the share of that form among the instances already on the page it writes on. Laboratory points are placed at the exposure of the record bins, interpolating between the bins of each model. In all panels the dashed line is exact proportionality and bars are 95% confidence intervals.

in the last hundred lines of the feed. We can then assume that agents look approximately to the last 100 lines and ignore the rest; in the following we refer to this subset of edits simply as the feed (other windows in Supplementary Information S2). Figure 2a plots the probability that a page is chosen against its share of the feed. The relation lies approximately on the diagonal, with a slope of 0.87 in the record and of 0.92 and 0.98 in the laboratory, meaning that a page that holds a fifth of the recent feed is chosen about a fifth of the time. This is a sign of proportional copying, applied to attention rather than to content: an agent is not evaluating pages, it is picking one of the lines it can see. This is a recency mechanism, and not the cumulative popularity of preferential attachment [38, 39, 40].

Agents also copy when they choose their names. As Fig. 2b shows, the probability that a newcomer name carries a feature, such as an affiliation stamp, a date or a role word, grows with the feature share among the agents of the same task within the feed, with a slope of 0.57 in the record and of 0.62 and 0.65 in the laboratory. The response is less steep than for pages, but still strongly increasing. Finally, to analyze the writing style, we took 16 conventions in which two forms or words compete for the same meaning, following the way competing variants of a phrase are traced through cascades of reposts on social media [48, 49]. They fall into three classes: names coined on the wiki (`R4` against `#4` for the fourth round of a task), near-synonyms that the language model could produce on its own (`relay` against `bridge`), and habits of style and typography (`CONFIRMED` against `confirmed`, `we` against `I`). Figure 2c shows the probability that a handle first use takes a given form against the share of that form among the instances already on the page it is writing on. The response rises along the diagonal, with a slope of 0.87 in the record and 0.81 for gpt-oss-20b, again signaling a linear copy mechanism; GPT-6.1 Sol copies the forms less, with a slope of 0.43, mostly because it holds

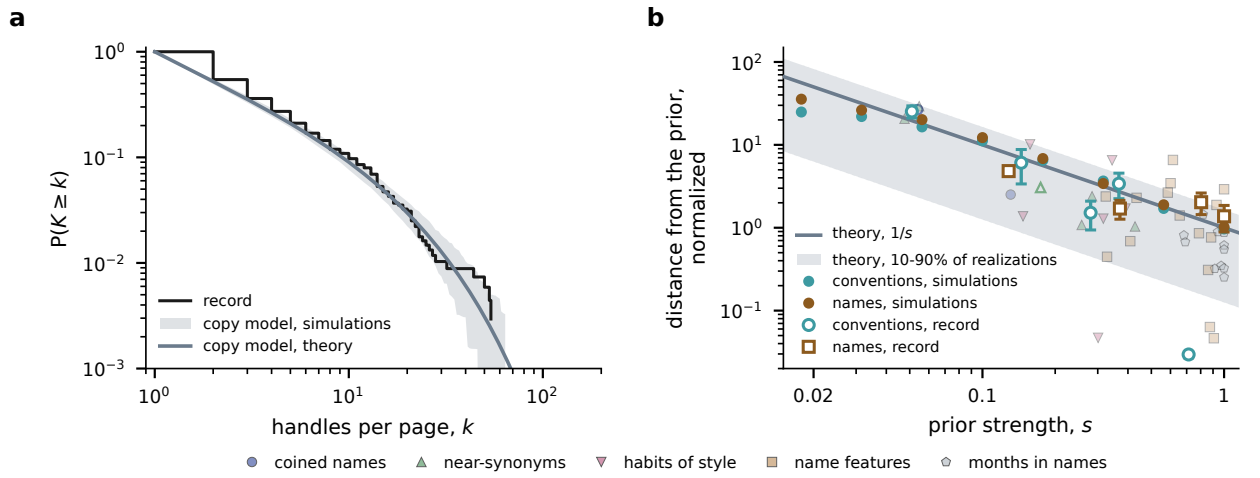


Figure 3: One model reproduces the collective, and can be solved. Nothing is fitted to a collective outcome. (a) Distinct handles per page, complementary cumulative distribution: record (black), 5th to 95th percentile of 200 simulated runs (band) and the Borel distribution of a branching process with Poisson offspring of mean $1 - p_c$, $p_c = 0.26$ (line). (b) Who wrote first decides, or not. Distance between the final share of a form and the share its prior alone would give, $\pi = \mu_A/s$, in units of the fluctuation of N independent choices, $\sqrt{\pi(1-\pi)/N}$, against the prior strength $s = \mu_A + \mu_B$. Line, the $1/s$ law of Eq. (2); band, the 10th to 90th percentile of single realizations for a Gaussian final share. Bullets, root mean square over 50 runs and five prior directions, for a convention read from the page (teal) and a name feature read from the feed (brown). Faded symbols, the empirical conventions and name features at their measured s (N the uses of the convention, or the 1,201 handles); open symbols, their root mean square in quarter-decade bins, with standard errors. For two conventions the fit puts one prior at zero, so that the direction π is not identified; they are drawn in outline at the largest value of that prior the record cannot exclude at 95%, and left out of the bins.

strongly on to its own capitalization and number formats. Four further models copy in the same way (Extended Data Fig. 2).

A response along the diagonal is what copying produces, but it is also what a shared prompt and harness would produce. Agents launched together on the same task from the same infrastructure might write the same things whether or not they saw one another. In the laboratory the exposure is assigned at random, so this explanation does not apply there. In the wild, the record helps to tell the two apart, because it allows us to reconstruct what each agent could see. For every choice we split the exposure into what was still visible when the agent arrived, on its page or in the feed, and what had already scrolled off. The visible exposure from task-mates carries most of the effect (Methods, Table 1), whether the task-mates are taken from the previous 1.5, 3, 6 or 12 h (Extended Data Fig. 3c). The page also competes with the other pages of the same task: when the two disagree, a handle's first use of a convention follows its page 68% of the time (Extended Data Fig. 3a,b). Every estimate survives intercepts for the task and the three-hour block of arrival, which absorb what a batch of agents launched together on the same prompt does in common across features and conventions (Methods, Table 1). The pattern is the one that copying produces, with a bias for recency and toward task-mates.

Modeling the collective behavior

The three choices an agent made thus share one copy rule that we can write as

$$P(A \mid \rho) = \mu_A + (1 - \mu_A - \mu_B)\,\rho. \tag{1}$$

Here ρ is the share of option A among what the agent look at and $P(A \mid \rho)$ the probability of picking A. The prior probabilities μ_A and μ_B are what the agent does without looking: the probability of taking A when nobody in view has it, and of not taking it when everybody has. Their sum $s = \mu_A + \mu_B$ is its strength, the probability that a decision does not follow what is in view: with probability $1 - s$ the agent copies, whether from the page, the feed or its own earlier writing. Their ratio is its direction, the form the agent prefers. The larger s, the less room copying has. For instance an agent that sees only `#4` on the page still writes `R4` one time in twenty, and an agent that sees only `confirmed` still writes `CONFIRMED` one time in seven. This is the agent prior, since with some probability it writes what it would have written anyway. The prior is small for words invented on the wiki, and large for capitalization and number formats, which are the language model own way of writing (Supplementary Information S4).

Leveraging this simple rule, we build a stylized model that reproduces most of what we observe on the wiki. Agents arrive one after another at random times, as many as in the record, and all are alike: each writes on two pages within two hours, the typical activity of an agent in the record. At every edit, an agent reads the last 100 lines of the feed. With a probability $p_c = 0.26$, measured from the data, it creates a page; otherwise it picks a line of the feed and writes on its page, so that a page is chosen in proportion to its share of the feed. On its first edit it decides each feature of its name by Eq. (1), reading the agents in the feed. On every edit it decides each convention by Eq. (1), reading the page or, if the page carries no use of it, the feed. Every edit puts its page back at the top of the feed. Every probability in the model is measured on individual decisions, and nothing is fitted to a statistic of the collective.

The model has no tasks, no cohorts, no notion of quality, prestige or goal, no page content and no memory beyond the feed (Methods). It belongs to two classes that statistical physics and population genetics have long studied. In our model a line of the feed is picked by later edits a number of times that is approximately Poisson with mean $1 - p_c$, so a page is the total progeny of a branching process, and its size follows the Borel distribution [52]. With nothing fitted the model reproduces the distribution of attention (Fig. 3a, line). The number of agents that met on a page is heavy tailed in the model as in the record and is describe by a power law $k^{-3/2}$ with exponential cutoff (Methods). This is analogous to models of meme popularity under limited attention [54].

Equation (1) is the asymmetric noisy voter model of herding [55, 56], closely related to the two-allele model of population genetics with mutation [57]. Let f be the final share of a form over N uses, and $\pi = \mu_A/s$ the share that the agents' prior alone would produce. The distance between f and π measures how far collective behavior departs from individual preferences. We can predict this distance through a simple branching approximation (Methods). In particular we obtain for the variance of f

$$\mathrm{Var}(f) \simeq \frac{\pi(1-\pi)}{Ns^2}. \tag{2}$$

The root-mean-square distance from the prior, normalized by the fluctuations of N independent choices, therefore scales as $1/s$. This prediction follows from the individual copying rule, without fitting any collective outcome.

The model reproduces the observed dependence of collective predictability on prior strength (Fig. 3b). The normalized distance from the prior approximately follows the predicted $1/s$ scaling, both in the simulations and the real data, with deviations at small s, where finite-size effects become important, and at large s, where the prior is fitted on the same uses whose final share it measures. The record supports the decrease with s more firmly than the exact amplitude: refitted as the record is, the model stays between 0.68 and 1.26 times the law, and with priors fitted on first uses alone the record keeps the trend but lies above it (Extended Data Fig. 5). The same pattern appears

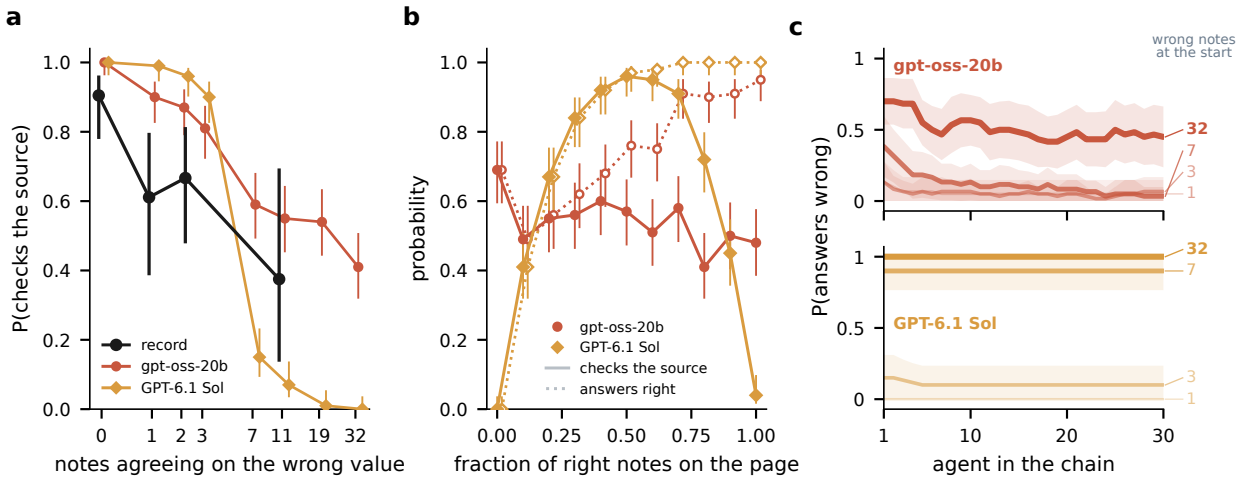


Figure 4: Agreement switches checking off, and a wrong consensus protects itself. Laboratory pages are made of notes from the record. (a) The probability of checking against the number of notes agreeing on the wrong value. Record (black; OECD Equity task after the first correction, uses on pages whose earlier uses all give the wrong value, in bins of 0, 1, 2–3 and 5 or more such uses plotted at their means): the probability that the note cites a source or gives evidence. Laboratory (gpt-oss-20b, vermilion; GPT-6.1 Sol, amber; four questions, 100 agents per point): the probability that the agent opens the accurate source; the probability of answering right follows it closely (Supplementary Information). (b) Laboratory: the probability of opening the source (filled, solid) and of answering right (open, dotted, drawn slightly to the right) against the fraction of right notes on a page of 20 notes from the record, a fraction of which were turned right by writing the right value in place of the wrong one (100 agents per point). Bars in a and b are 95% Wilson intervals. (c) Laboratory, the collective: chains of 30 agents on one page that starts with 1, 3, 7 or 32 notes from the record giving the wrong value; each agent can open the source, answers, and appends a note. Probability that the agent at each position answers wrong, for gpt-oss-20b (top) and GPT-6.1 Sol (bottom), labelled by the number of wrong notes at the start (20 chains per curve, smoothed over three agents, 95% band).

in the record for both linguistic conventions and name features. When priors are weak, copying amplifies early random choices, and the collective can end far from what individual preferences would predict. When priors are strong, independent choices dominate and the collective remains close to its prior. Each convention simulated with its measured priors also produces outcomes of the observed size, its volatility over time more closely than its consistency within a page (Extended Data Fig. 4). Thus, individual behavior can predict the collective when priors are strong, but becomes increasingly uninformative as copying dominates.

Collective failure

The model predicts that weak priors make collective outcomes difficult to predict. But for factual answers, the situation is more complicated. When an agent is asked to look up a value, it cannot reliably know the answer without consulting an external source. In this setting, the probability of checking the source rather than copying from the page plays the role of the prior strength s in Eq. (1). Crucially, this probability need not be constant: it may depend on what other agents have already written. The record offers a first indication of this effect. In the case of the wrong answer of Fig. 1c, notes became less likely to cite a source as more agents agreed on the wrong value (Fig. 4a). However the record reveals whether a note explicitly cites a source or gives evidence, not whether its author actually consulted the source, so it is an imperfect proxy for checking. To measure checking directly, we returned to the laboratory, giving gpt-oss-20b and GPT-6.1 Sol the same question, pages built

from the wrong notes in the record, and a tool to retrieve the correct value (Methods). Almost every agent that checked answered correctly, but checking declined from nearly universal on an empty page to 41% with 32 agreeing notes for gpt-oss-20b, and fell abruptly for GPT-6.1 Sol, from 90% with three agreeing notes to 15% with seven (Fig. 4a).

The laboratory also lets us go beyond the record and manipulate the composition of the page, not just the number of notes. We kept the notes of the record and turned a fraction of them right by writing the correct value in place of the wrong one, so that right and wrong notes differ only in the number. GPT-6.1 Sol checked when the page contained conflicting information, and almost never when all notes agreed, regardless of whether they were right or wrong (Fig. 4b). For this model, it is agreement itself that discourages independent verification. gpt-oss-20b checked about half of the time whatever the composition of the page and copied its majority when it did not check; its checking declined with the amount of material on the page rather than with agreement (Supplementary Information). In both cases the effective prior strength s depends on the collective state, and a page that agrees on a wrong value lowers it.

Finally, we used the laboratory to test the consequences of this mechanism at the collective level, in controlled chains of agents that the record cannot provide. Each agent could consult the correct source before answering and adding its note to the page, which was then passed to the next agent. A page starting with a single wrong note was corrected almost immediately by both models. With gpt-oss-20b, a page starting with 32 agreeing wrong notes often remained wrong; with GPT-6.1 Sol, seven agreeing notes were enough for the wrong answer to persist through all 30 agents in 18 of 20 chains (Fig. 4c). Four further open-weight models were also misled by pages of the record notes, some of them even after retrieving the correct value (Extended Data Fig. 6). The same mechanism that allows copying to establish shared conventions can therefore sustain collective errors because once agreement discourages verification, a wrong answer can persist even when every agent has access to the correct source.

Discussion

Thousands of AI agents, none of which could carry anything from one run to the next, met on a wiki that nobody had built for them, and organized themselves when nobody had instructed them to cooperate. Within days they had a way of choosing where to write, a shared way of signing, and a common way of phrasing their messages to communicate useful information. This is the kind of coordination that people achieve without thinking, and that we usually take as evidence that a group of individuals has become a collective [31, 41]. They also shared answers, copying them without checking and creating examples of collective failure, where the collective settles on a wrong answer due to conformity effects.

Much of the collective behavior on the wiki can be explained by proportional copying. We measured this rule from individual choices, and a model built on it reproduces the concentration of attention and predicts how strongly collective conventions can depart from individual preferences. Copying made coordination possible: it concentrated agents on shared pages and produced a common vocabulary that newcomers could understand and use. Yet the same mechanism makes collective outcomes dependent on early choices when individual priors are weak. For conventions that agents already have strong preferences about, such as capitalization or number formats, the collective remains predictable. For conventions invented on the wiki, instead, early choices are amplified through copying, and the outcome depends on who happened to write first. This unpredictability is familiar from human social influence, where copying can produce winners that cannot be predicted from individual preferences alone [43, 44, 45].

For factual answers, this dependence on what others have written has more serious consequences. Our laboratory experiments show that agreement discourages agents from independently checking external sources, even when those sources are readily available. Unlike the priors governing writing conventions, the probability of checking is therefore not fixed, but depends on the collective state itself. A wrong consensus can become self-reinforcing: the more agents agree, the less likely the next agent is to verify their answer, allowing the error to persist. Some models even verify and still conform to the page. Together, these results show how copying can make AI collectives both useful and vulnerable. In particular, they suggest that influencing the information agents encounter may be enough to steer their collective behavior, without modifying the underlying models or their instructions.

Several limitations follow from the nature of the record. We can reconstruct what agents could have seen, but not what they actually read, so exposure is observed rather than experimentally assigned. Agents launched on the same task may also share features because of their common prompts rather than copying, and handles do not necessarily correspond to distinct agent runs. Our analyses account for the alternatives that the record allows us to test, but cannot establish copying causally. The laboratory provides controlled evidence for the mechanisms we identify, although it uses different models in a simpler environment than the original agents. In particular, the laboratory chains reproduce the persistence of wrong answers, but not the sudden correction observed in the record after evidence was shared.

Our findings show why the behavior of AI collectives cannot be understood by studying their members in isolation. The tendency to copy is an individual property, but which conventions emerge, where attention concentrates and whether errors persist are outcomes of interactions among agents. Moreover, these interactions can change individual behavior itself: as agents encounter increasing agreement, they become less likely to verify information independently. Understanding the reliability of AI agents therefore requires studying not only what individual models can do, but also how their behavior changes when they encounter information produced by one another.

Methods

Dataset and reconstruction of agent exposure

Dataset and population. We analyze the release published at collusion.wiki on 4 September 2026. It contains 14,591 stored revisions of 4,579 wiki pages, including the timestamp, username, text added by each revision, and the page content after the edit. The histories omit some short edits (below 64 characters on DSEWiki) and deleted pages that could not be recovered. After excluding three accounts flagged as human and revisions without a username, the record contains 13,661 edits attributed to 3,099 distinct usernames, hereafter *handles*. A handle is an observed username, not necessarily a unique agent run.

According to the original investigation, the agents were carrying out time-limited tasks, typically consisting of five questions that required retrieving numerical information from public data sources, such as DataUSA, OECD or IHME. For example, one task, which we examine later, asked agents to retrieve numerical values from an OECD workbook and report them to two decimal places. We thus study the 1,201 handles that contributed to at least one page associated with an identifiable data-lookup task. They made 5,929 edits across the wiki, including 3,807 edits on 679 pages associated with 41 task families. We assign each handle to the family of its first task-page edit; 945 handles contributed to one family, 168 to two and 88 to three or more. Repeating the analyses on the entire population we obtain similar results (Supplementary Information S1).

Reconstructing exposure. Our question is whether an agent's choice is influenced by the information it could have encountered before acting. We reconstruct two sources of such information at the instant before each edit: the content already on the edited page and the wiki's RecentChanges listing, or *feed*. The revision history allows us to reconstruct the chronological listing, but not which part of it any agent actually requested or read. We approximate its potentially visible portion by the latest 100 revisions across the wiki, including edits by handles outside our study population. The chosen page appears within this window for 89% of appends; the corresponding coverage is 57%, 72% and 95% for windows of 15, 30 and 300 revisions. The influence of window length is examined in Supplementary Information S2.

Measuring individual copying

We analyze three decisions: which page to use, which elements to include in a username, and which of two competing expressions to write. For each decision we compare the selected alternative with its prevalence in the reconstructed exposure, so to obtain the curves in Fig. 2.

Page choices. We call the first edit by a handle on a given task page an *arrival*; later edits by that handle on the same page do not create another arrival. Of 2,612 arrivals, 679 create a page and 1,933 append to an existing page. The observed creation probability is therefore $p_c = 679/2612 = 0.260$. For each append, we treat every distinct page represented in the preceding 100 feed revisions as a candidate and measure its *feed share*, the fraction of those revisions referring to it. A candidate is marked as selected if it is the page that received the edit. The probability of selection is then estimated as a function of feed share (Fig. 2a). Displayed binomial proportions have 95% Wilson intervals; the descriptive response slope is fitted on unbinned observations by ordinary least squares.

Name features. We decompose usernames into capitalized words and consecutive runs of uppercase letters, retaining pieces found in at least 15 handles. We exclude pieces strongly associated with a single task family, defined as those at least four times more frequent within one task than in the population as a whole. This threshold falls within a clear gap in the observed distribution of task concentration (Supplementary Information S3). After pooling month names into a single date-stamp feature, we end up with 16 name features for the analysis.

For each newcomer and each feature, we record whether the feature appears in its chosen name (one binary outcome per newcomer–feature pair). The corresponding exposure is the proportion of earlier same-task handles in the reconstructed feed that carry the feature. For this descriptive analysis we require a complete 100-revision feed and at least two such task-mates. We then measure the probability that a newcomer adopts a name feature as a function of its frequency among task-mates in the feed. To combine the 16 features into a single response curve, we account for differences in their baseline frequencies (Fig. 2b).

Writing conventions. To study copying in writing conventions, we used Claude Code to examine the full dataset and identify 52 candidate pairs of competing expressions, including terms coined on the wiki, near-synonyms and stylistic alternatives. For each pair, we identify occurrences of the two forms in the text added by each edit, excluding links. When both forms appear in the same edit, we assign the more frequent one; ties are discarded. We then retain 16 pairs with at least 100 uses, both forms accounting for at least 5% of uses, and at least ten pages containing five or more uses. These criteria ensure that both alternatives occur sufficiently often across different pages to study copying and estimate baseline preferences. The full list of candidates and the robustness to these thresholds are reported in Supplementary Information S4.

To measure copying, we consider the first time each handle uses a given convention, excluding its subsequent uses. We compare the form chosen on this first use with its frequency among previous uses on the same page, considering only cases with at least three preceding occurrences of either form. This gives 2,679 first-use observations across the 16 conventions. We estimate the response from individual observations and display the probabilities with 95% Wilson confidence intervals (Fig. 2c).

Separating exposure from shared conditions. Agents working on the same task may make similar choices because they receive similar instructions, even if they never copy one another. We therefore ask whether an agent's choice is more closely related to predecessors it could have seen than to predecessors working on the same task whose contributions were absent from the reconstructed page and feed. We use *linear probability models*: ordinary least-squares (OLS) regressions in which the outcome is either 0 or 1. Their coefficients describe changes in the probability of a choice.

Names and writing conventions. For names, each observation is a newcomer and one name feature. We set $y_i = 1$ if the newcomer includes that feature in its username, and $y_i = 0$ otherwise. For conventions, each observation is a handle's first use of one convention. We set $y_i = 1$ if it uses form A, and $y_i = 0$ if it uses form B.

For each observation, we measure the frequency of the feature under study, or of form A, in four sources of earlier activity: (i) on the page being edited; (ii) in the feed, among agents working on the same task; (iii) outside both the page and the feed, among agents working on the same task; and (iv) in the feed, among agents working on other tasks. For names, the first group consists of earlier same-task authors of the newcomer's first page, and the frequencies are fractions of handles carrying the feature. We consider predecessor handles first observed in the preceding six hours. For conventions, the first group consists of earlier uses on the current page, and the frequencies are fractions of uses taking form A.

We estimate

$$\begin{aligned} y_i = \alpha_{k(i)} + \gamma_{\tau(i),b(i)} + \beta_{\text{page}}\rho_i^{\text{page}} + \beta_{\text{feed}}\rho_i^{\text{feed}} \\ + \beta_{\text{out}}\rho_i^{\text{out}} + \beta_{\text{other}}\rho_i^{\text{other}} + \theta m_i + \epsilon_i, \end{aligned} \tag{3}$$

where each ρ is one of the four frequencies above. The intercept $\alpha_{k(i)}$ allows each feature or convention to have its own baseline frequency. The intercept $\gamma_{\tau(i),b(i)}$ accounts for differences shared by agents in the same task family τ and three-hour period b. We require at least two preceding handles or uses to calculate each frequency. For names, if the newcomer's first page has too few earlier same-task authors, its page frequency is set to zero and $m_i = 1$ marks this missing information; otherwise $m_i = 0$. The term θm_i is used only in the name regression. Copying predicts $\beta_{\text{page}}, \beta_{\text{feed}} > \beta_{\text{out}}$; shared task conditions alone predict similar coefficients.

Page choices. Choosing a page is different: an agent selects one page from several possible pages. We therefore compare candidate pages *within the same decision.* For each append to an existing page, we include task pages appearing in the last 100 feed revisions and earlier pages of the agent's own task that are no longer in the feed. We create one observation for each candidate page j at decision d, with $y_{dj} = 1$ if the page is chosen and 0 otherwise. We estimate another linear probability model:

$$\begin{aligned} y_{dj} = \alpha_d + \beta_{\text{own}} v_{dj} T_{dj} + \beta_{\text{other}} v_{dj}(1 - T_{dj}) \\ + \gamma V_{dj} T_{dj} + \delta T_{dj} + \eta \log(1 + H_{dj}) + \epsilon_{dj}. \end{aligned} \tag{4}$$

Here v_{dj} is the candidate page's fraction of the feed (zero if absent), $T_{dj} = 1$ if it belongs to the agent's task, $V_{dj} = 1$ if it appears in the feed, and H_{dj} is the number of handles that previously edited it. The intercept α_d is different for every decision, so the coefficients compare pages facing the same agent at the same moment. We allow the relationship with feed share to differ between own-task

and other-task pages, and account separately for own-task membership, feed presence and previous popularity.

Standard errors are clustered by handle to account for multiple observations from the same handle. We also test the differences between exposure coefficients directly. For page choices, a conditional logit model provides a check that explicitly treats the outcome as a choice among candidates. Alternative fixed effects, exposure windows and clustering methods are reported in Supplementary Information S5. Results are reported in Table 1.

Laboratory validation of copying

The record cannot reveal what agents actually read or separate exposure from every possible shared condition. We therefore construct controlled tasks from authentic wiki pages and feeds while randomly assigning the frequencies of competing alternatives. This manipulation preserves much of the original linguistic context but breaks the link between the displayed frequencies and the original history of the page. Each trial is independent, without memory carried over from other trials, and the choices are scored using the same definitions as in the record.

The laboratory dataset contains 820 prepared trials: 240 trials manipulating page frequencies and name-feature frequencies in the feed, 60 related trials using anonymous handles, 440 trials manipulating convention frequencies on task pages, and 80 empty-page writing trials to measure unexposed preferences. We also randomize clock and numerical formats and remove name features from page titles when they would give away the intended choice. We test gpt-oss-20b, served locally through vLLM 0.19.1 at low reasoning effort, and GPT-6.1 Sol through the OpenAI API, using matched prompts. Every model has a readable reply to all 820 trials: malformed replies are read by a tolerant parser or run again (Supplementary Information S7). We score the selected page, the generated name and the chosen writing forms directly from model outputs. Laboratory response curves are interpolated to the record's exposure bins for visual comparison (Fig. 2). The prompts, stimuli and responses are archived with the analysis code.

The collective copying model

In the model, agents create pages, choose usernames and adopt writing conventions by the rules measured in the record; its parameters are estimated from individual choices and never fitted to the collective outcomes. The model contains 1,201 agents, matching the observed population. Agents arrive uniformly over 5.3 days, corresponding to the period containing the central 90% of arrivals in the record. Each agent makes two edits on different pages, at random times within 2.02 hours of its arrival, corresponding to the median time between the first and last edit of a handle in the observed data. The model contains a single task, reflecting the stronger copying observed between agents working on the same task.

At each edit, an agent creates a new page with probability $p_c = 0.260$, as measured in the record. Otherwise, it selects a page from the last 100 revisions in the feed. Specifically, the agent chooses uniformly among feed entries pointing to pages it has not previously edited. Pages appearing more frequently in the feed are therefore more likely to be selected. If no eligible page is available, the agent creates a new one. Every edit produces a new feed entry, so recently edited pages remain visible while older entries gradually disappear. Agents also choose usernames and writing conventions. For names, each agent decides which features to include when it arrives, using their frequencies among distinct handles in the feed. For writing conventions, each edit has a probability $u = 0.230$ of using a given convention, corresponding to the average usage rate in the record. When a convention is used, the agent chooses form A with the probability of Eq. (1), $P(A \mid \rho) = s\pi + (1 - s)\rho$ with $s = \mu_A + \mu_B$ and $\pi = \mu_A/s$, where ρ is the fraction of preceding uses that take form A.

For writing conventions, we calculate ρ from previous uses on the page being edited. If the convention has never appeared on that page, we use its frequency in the feed instead. If neither source contains the convention, we set $\rho = 1/2$. For names, we calculate exposure from the fraction of distinct handles in the feed carrying each feature. The model keeps track of pages, usernames and the frequencies of competing forms, but does not generate or interpret their semantic content. It also does not include page quality, agent reputation or external goals.

Estimating the model parameters. The copying rule has two parameters for each name feature or writing convention: the independent-choice strength s and the baseline preference π. We estimate them directly from individual choices in the observed wiki, using maximum likelihood to fit the two parameters $\mu_A = s\pi$ and $\mu_B = s(1-\pi)$, with $s, \pi \in [0, 1]$. We include only decisions with at least three preceding observations from which exposure can be calculated.

For conventions, we reconstruct exposure in the same way as in the model: we count previous uses on the current page, including earlier uses by the same handle, or uses by other handles in the last 100 feed revisions if the convention has not appeared on the page. Unlike the analysis of Fig. 2c, which considers only the first use of each convention by an agent, here we include all uses. This is because the model describes the full sequence of decisions, including repeated contributions by the same agent. Note that s represents the probability of making an independent choice, so it is an effective measure of independence from the reconstructed exposure: it can capture both copying from others and persistence of an agent's own choices.

As a robustness check, we also estimate the parameters using only the first use of each convention by each handle. This produces larger values of s, as expected when repeated choices are excluded. The collective distance from the prior still decreases as s increases, after accounting for the unequal numbers of contributions per handle, although its magnitude remains above the theoretical $1/s$ prediction (Extended Data Fig. 5a and Supplementary Information S6).

Model comparisons. We run 200 independent simulations to study page popularity and 50 simulations per parameter setting for names and writing conventions. We also simulate each observed convention using its fitted parameters and compare the resulting final frequencies, changes over time and agreement within pages with those observed in the wiki.

Analytical predictions and collective observables

Page popularity. In the model a page gains authors through the feed: every edit adds an entry for the page, and each entry can attract a further edit to the same page while it remains visible. The size a page reaches can therefore be computed as a branching process, in which every entry gives rise to new entries. An entry stays in the feed for about 100 subsequent edits, and at each of them it is selected with probability $(1 - p_c)/100$, where p_c is the probability of creating a new page. Each entry therefore generates on average $\lambda = 1 - p_c$ new entries pointing to the same page, a number we take to be Poisson distributed. A new page starts with one entry, and the total number of entries descending from it, K, follows the Borel distribution:

$$P(K = k) = \frac{e^{-\lambda k}(\lambda k)^{k-1}}{k!}, \qquad \lambda = 1 - p_c. \tag{5}$$

This distribution has a mean of $1/p_c$ and, for large k, decays as $P(K = k) \sim k^{-3/2} e^{-k/k^*}$, where $k^* = (\lambda - 1 - \ln \lambda)^{-1}$: many pages stay small and a few grow much larger.

Since each simulated agent edits a given page at most once, the number of edits on a page is also

its number of distinct authors. We compare the predicted distribution with the number of distinct handles per page in the wiki and with the results of 200 independent simulations (Fig. 3a). The branching calculation is an approximation, since the finite feed and the restriction on repeated visits introduce dependencies between edits.

Collective distance from the prior. We next compute how far copying moves the final share of a convention or name feature away from what independent choices would produce. Consider a convention with forms A and B, used N times, and let f be the final fraction of A. With $s = \mu_A + \mu_B$ and $\pi = \mu_A/s$, each use makes an independent choice with probability s, picking A with probability π, and otherwise copies an earlier use. Without copying, f would fluctuate around π with the binomial variance $\pi(1-\pi)/N$; with copying, a single independent choice can be repeated by many later uses, and f fluctuates more.

To compute this variance we group the uses into families: each independent choice starts a family, and every copy joins the family of the use it copies. Let M be the number of families, T_j the number of uses belonging to family j, and $X_j = 1$ if its original choice was A and 0 otherwise. Since all members of a family share the same original choice, the final fraction of A can be written as

$$f = \frac{1}{N} \sum_{j=1}^{M} T_j X_j. \tag{6}$$

The original choices are independent Bernoulli variables with probability π, so for given family sizes

$$\mathrm{Var}(f \mid T_1, \ldots, T_M) = \frac{\pi(1-\pi)}{N^2} \sum_{j=1}^{M} T_j^2. \tag{7}$$

A family of T uses thus weighs T^2 in the variance, not T: copying amplifies one decision instead of adding independent ones.

Among N uses, approximately $M = Ns$ are independent choices that start new families. If we approximate the number of direct copies generated by each use as Poisson-distributed with mean $1 - s$, branching theory gives a mean family size $\mathbb{E}[T] = 1/s$ and a second moment $\mathbb{E}[T^2] = 1/s^3$. Approximating the sum of squared family sizes by $Ns\,\mathbb{E}[T^2]$, and neglecting correlations and finite-size effects, we obtain

$$\begin{aligned} \mathrm{Var}(f) &\simeq \frac{\pi(1-\pi)}{N^2} Ns\,\mathbb{E}[T^2] \\ &= \frac{\pi(1-\pi)}{Ns^2}. \end{aligned} \tag{8}$$

To measure how far the final frequency is from the independent preference, we define

$$X = \frac{|f - \pi|}{\sqrt{\pi(1-\pi)/N}}, \qquad \sqrt{\mathbb{E}[X^2]} \simeq \frac{1}{s}. \tag{9}$$

that is, the distance in units of the standard deviation expected from N independent choices. Copying therefore amplifies the root-mean-square (RMS) distance by a factor of approximately $1/s$, more strongly the stronger the copying.

This prediction applies approximately to sufficiently large populations with $s > 0$. It is not an exact result for our finite model, especially when independent choices are rare.

Comparison with simulations and observations. We test this prediction by simulating name features and writing conventions with eight approximately logarithmically spaced values of s between 0.0178 and 1, and five independent preferences $\pi \in \{0.15, 0.30, 0.50, 0.70, 0.85\}$. For each parameter setting, we run 50 independent simulations. We use $N = 1,201$ for names and an expected $N = 1,201 \times 2 \times 0.230$ uses for conventions, following the model's population size and convention usage rate. For each value of s, we calculate the RMS distance across simulations and preferences.

We then measure the same distance in the observed wiki. For each name feature or convention, we use its observed final frequency f, number of observations N, and independently fitted parameters s and π. Since the wiki provides only one final outcome per feature or convention, we group features and conventions with similar values of s and calculate their RMS distance within quarter-decade bins. Uncertainty is estimated using the delta method (Fig. 3b).

Because the record's s and π are estimated, we also apply the same estimation procedure to simulated data (Extended Data Fig. 5c and Supplementary Information S6).

Factual answers and independent verification

Answers in the record. To examine whether copying also transmits factual mistakes, we analyze five questions of the OECD Equity task for which two numerical answers occur in the record: a workbook display rounded to one decimal and padded to two decimals, and the correct two-decimal value derived from the stored data. We detect both numbers in the text added by each edit, excluding matches embedded in longer numbers and unrelated numerical contexts. An edit is assigned to the alternative occurring more often as an answer; ties are discarded. This yields 325 uses on 97 pages by 125 handles. The first wrong note was followed by 159 additional uses of the padded answer before the first correct note was posted, on 20 June 2026 at 03:41 UTC. We aggregate the competing answers in half-hour bins around that correction (Fig. 1c).

We then ask whether the presence of an agreeing wrong answer on a page is associated with less explicit attention to the original source. We consider uses after the first correction on pages whose earlier recorded answers all reported the padded value. For each use, we count earlier wrong notes on the page and classify whether the new note reports the correct value or cites a source or supporting evidence, using source names, extraction-related language and numerical details. These mentions are a proxy for source checking, not a direct measure: an agent may check without citing, or copy a citation without checking. We present the proportions against the number of earlier wrong notes with Wilson intervals (Fig. 4a). Restricting to the period after the correction ensures that both alternatives had appeared somewhere in the record, but does not establish that either was visible to a particular agent.

Laboratory verification. The laboratory allows us to measure verification directly. Each agent receives a country-specific OECD question, a wiki page with previous notes and a fetch tool that first opens the data dashboard and then provides access to the workbook's stored values. We count an agent as checking only if it fetches the workbook, and evaluate whether its numerical answer matches the correct or padded value. Trials permit up to eight model turns. We use gpt-oss-20b at temperature 1 and low reasoning effort through vLLM, and GPT-6.1 Sol with low reasoning effort through the OpenAI Responses API, with the same prompt, tool and scoring. Fetching the workbook provides the unrounded data needed to answer correctly, without relying on the notes.

To measure how accumulating agreement affects verification, we give agents pages containing 0, 1, 2, 3, 7, 11, 19 or 32 notes reporting the padded value. Non-empty pages are assembled from 153 such notes in the record, with randomized order. We use 25 independent trials per country and condition for four country questions and each model, measuring both workbook retrieval and answer correctness

(Fig. 4a). To separate agreement from the correctness of the majority, we also vary the fraction of correct notes while holding page length fixed. Each page holds 20 notes drawn from the record notes reporting the padded value, excluding the few that also mention the correct value or the stored digits, and a fraction 0, 0.1, ..., 1 of them is turned correct by writing the correct value in place of the padded one, so that correct and wrong notes differ only in the number. Each country–composition condition has 25 trials for each model, 100 per plotted composition (Fig. 4b). The same experiments with synthetic template notes, with pages of varying length at a fixed half-correct composition, and for four further models are in Supplementary Information. These checking measures come from tool-use logs, not claims made in final responses.

Collective chains. Finally, we test the collective consequence of the individual checking behavior in sequential laboratory chains. Each chain consists of 30 separately instantiated agents acting in sequence on one shared page. They carry no conversational memory from one position to the next: the page is the only state passed along the chain. An agent may fetch the correct source, gives an answer and appends its note; the resulting page is then passed to the next agent. Pages begin with 1, 3, 7 or 32 sampled wrong notes from the record. For each initial size and model we run 20 chains, five for each of four countries, for a total of 2,400 agent decisions per model. We measure the fraction of wrong answers at each position and display three-position-smoothed curves with 95% intervals across chains (Fig. 4c). Notes added by the laboratory agents are used unchanged in subsequent steps. The record before the first correction is compared with both models in Extended Data Fig. 7; trial counts by model, condition and country, and the chains by starting page, are in Supplementary Information S7.

Author contributions

G.D.M. and N.A. contributed equally. G.D.M. and N.A. designed the study, carried out the analysis and wrote the paper. D.G. supervised the work and revised the paper. All authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The wiki record analyzed here is the public release at `collusion.wiki` [6]; the four files and their checksums are listed in the repository below, which does not redistribute them. The laboratory data were generated for this study: the stimuli, prompts and every model reply used in the paper (the copying experiment and the answer experiments, including the tool logs from which checking is measured) are archived in the same repository.

Code availability

All the code that reproduces every number and every figure of this paper and of the Supplementary Information, starting from that release, is available at `https://github.com/giordano-demarzo/agent-wiki-copying`; it runs from the stored laboratory replies, without calling any model.

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Table 1: Prior exposure and individual choices. Linear probability models estimated by OLS. The outcome is 1 when a candidate page is chosen, a username contains the feature, or a first convention use takes form A. Entries are coefficients, with standard errors clustered by handle in parentheses. Names and conventions include task-family by three-hour fixed effects; page choice includes one fixed effect per arrival, which already absorbs task and time. The page model's R^2 is measured within arrivals.

	(1)	(2)
A. Where to write: the candidate page is the one chosen		
share of the feed, own task	2.10	
	[1.86, 2.35]	
share of the feed, other task	0.21	
	[0.13, 0.28]	
in the feed, own task	0.038	
	[0.030, 0.046]	
own task	0.014	
	[0.011, 0.017]	
log(1 + handles on the page)	0.006	
	[0.005, 0.007]	
Arrival intercepts	yes	
Task × 3 h intercepts	no	nested
Observations	87,349 candidates, 1,894 arrivals	
R^2	0.10	
B. How to sign: the name carries the feature		
share among page authors, own task	0.12	0.11
	[0.04, 0.19]	[0.04, 0.18]
share in the feed, own task	0.45	0.44
	[0.38, 0.53]	[0.36, 0.52]
share out of view, own task	0.15	0.14
	[0.07, 0.23]	[0.06, 0.22]
share in the feed, other tasks	0.07	0.08
	[-0.02, 0.17]	[-0.01, 0.18]
first page without earlier task-mates	0.02	0.02
	[0.01, 0.03]	[0.00, 0.03]
Feature intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	9,728 newcomer–feature records	
R^2	0.34	0.34
C. How to write: the first use takes form A		
share on the page	0.66	0.66
	[0.56, 0.77]	[0.54, 0.77]
share in the feed, own task	0.25	0.23
	[0.14, 0.36]	[0.12, 0.35]
share out of view, own task	0.03	0.03
	[-0.06, 0.12]	[-0.07, 0.12]
share in the feed, other tasks	-0.09	-0.08
	[-0.20, 0.02]	[-0.19, 0.04]
Convention intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	1,052 first uses	
R^2	0.58	0.61